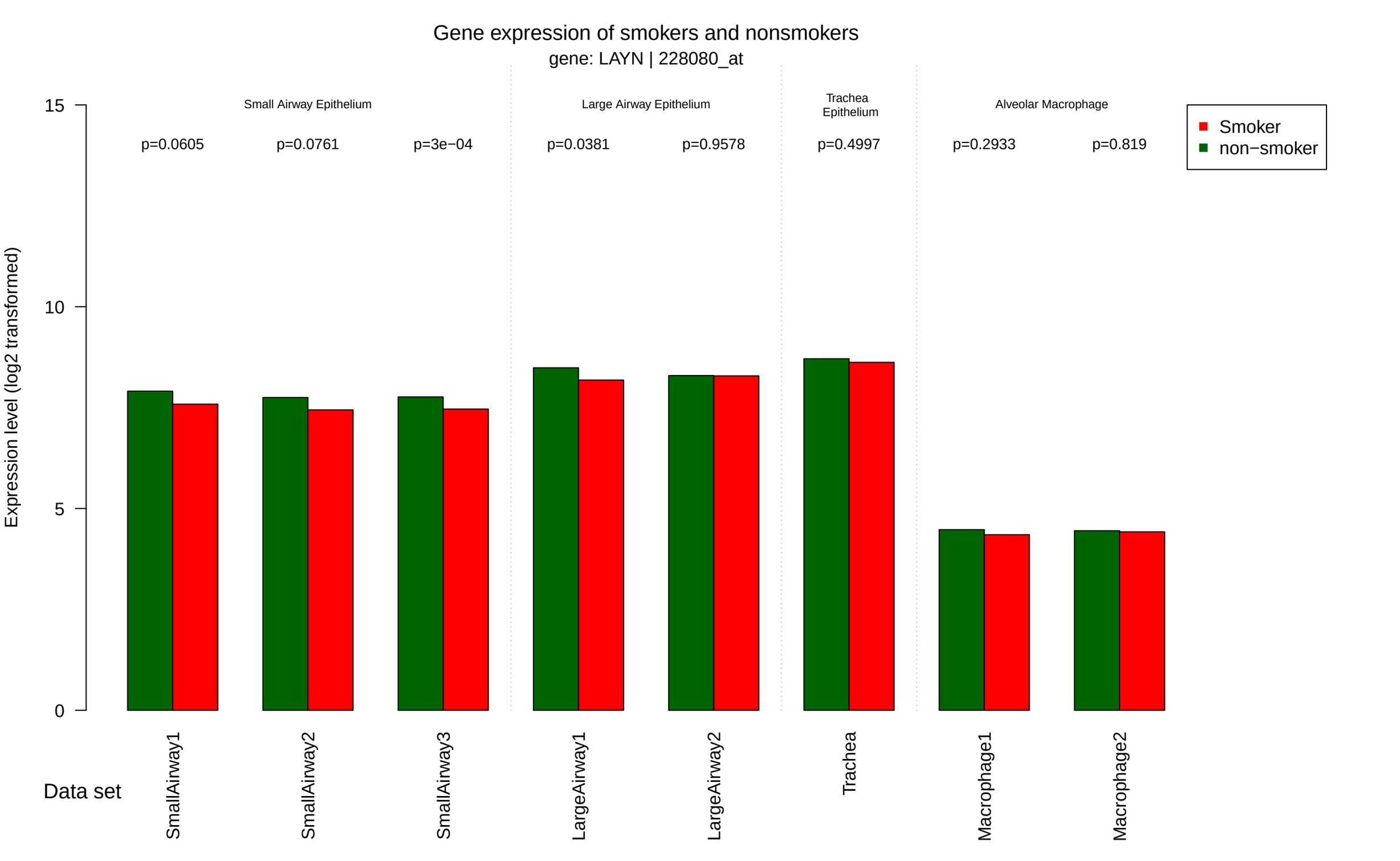
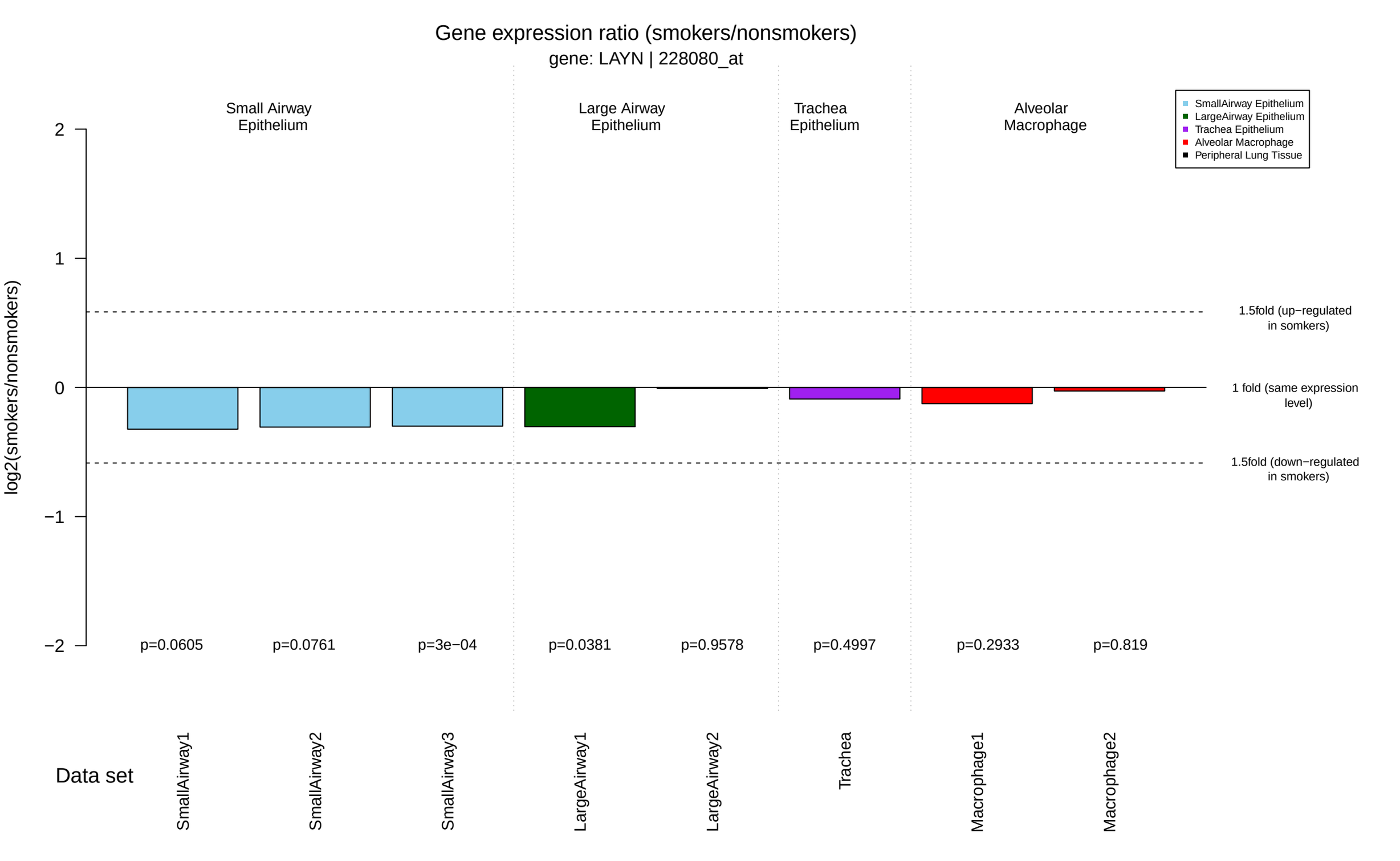
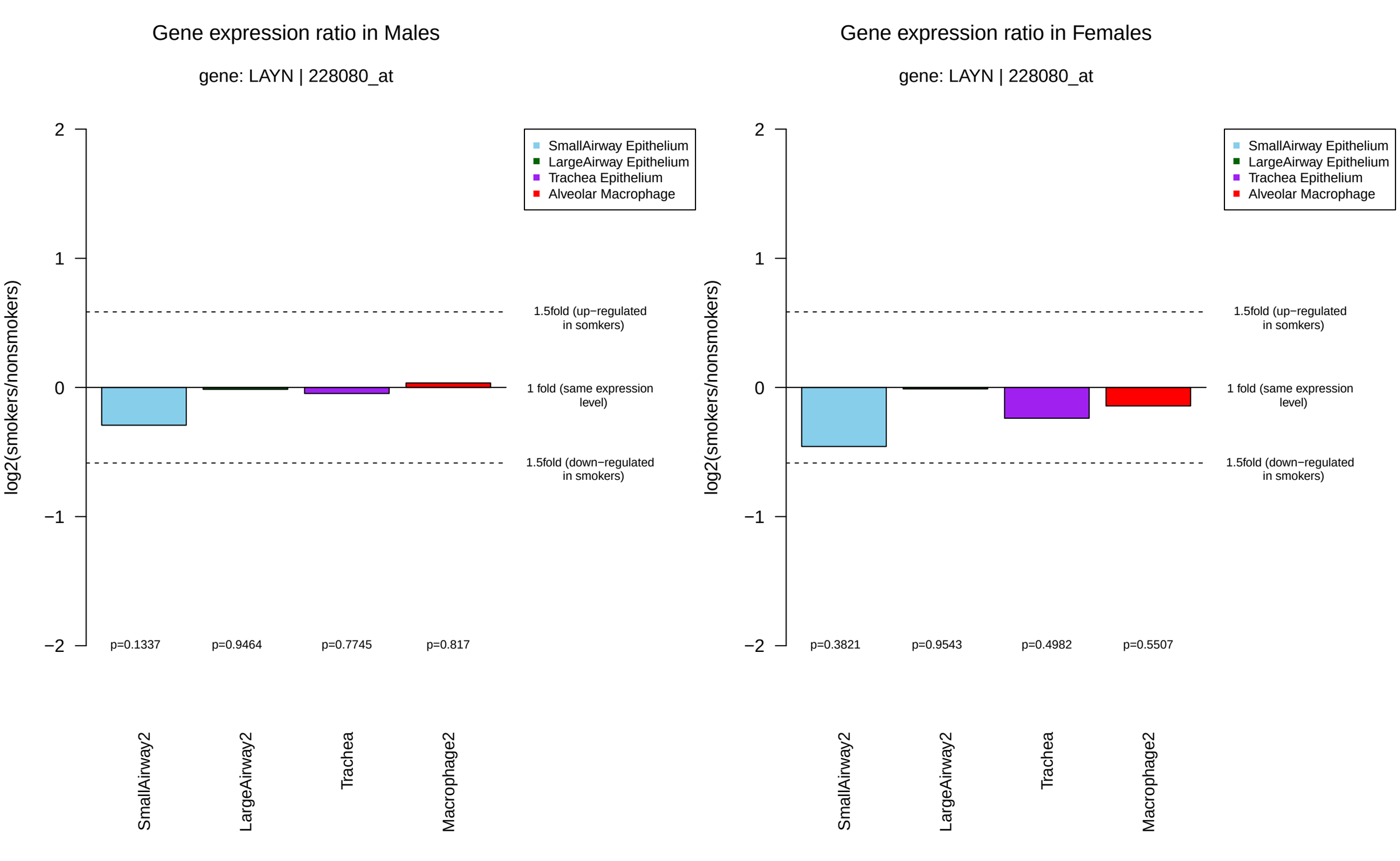
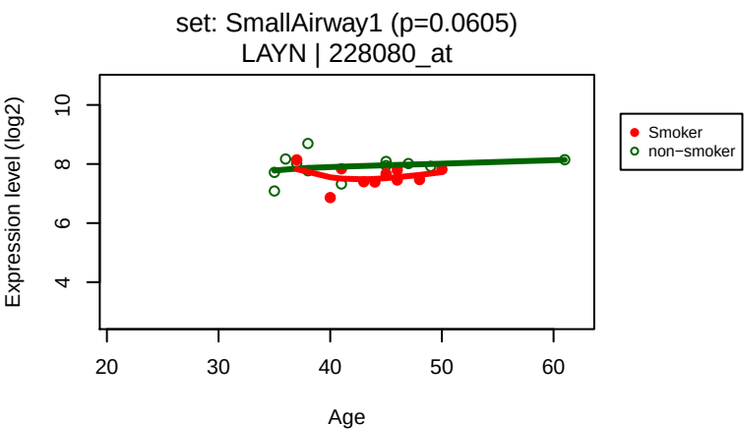
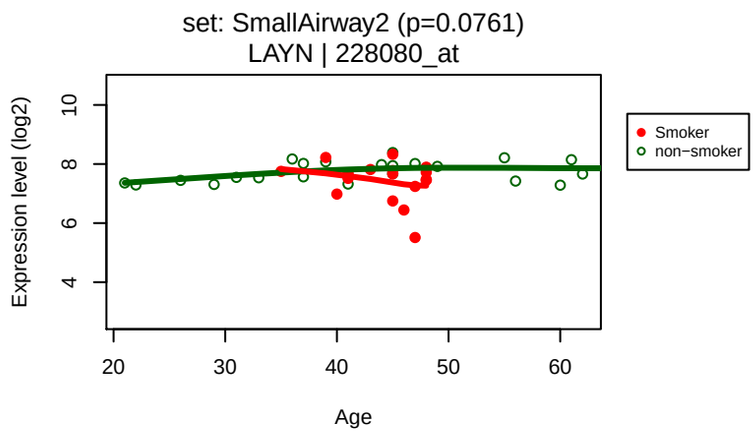
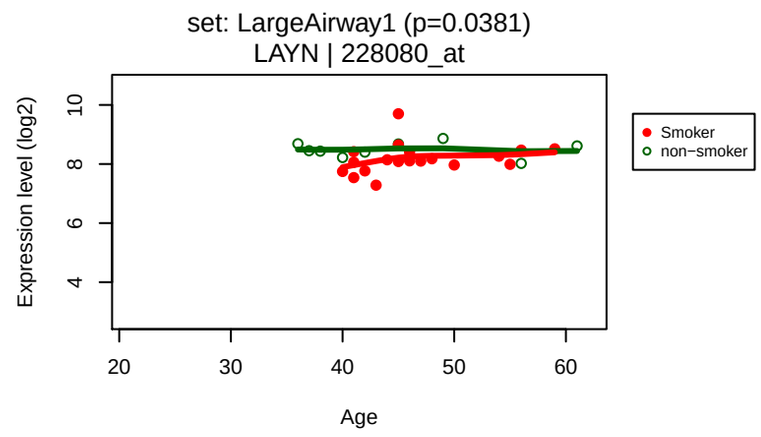

Gene expression of smokers and nonsmokers
gene: LAYN | 228080_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0605
p=0.0761
p=3e−04
p=0.0381
p=0.9578
p=0.4997
p=0.2933
p=0.819
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

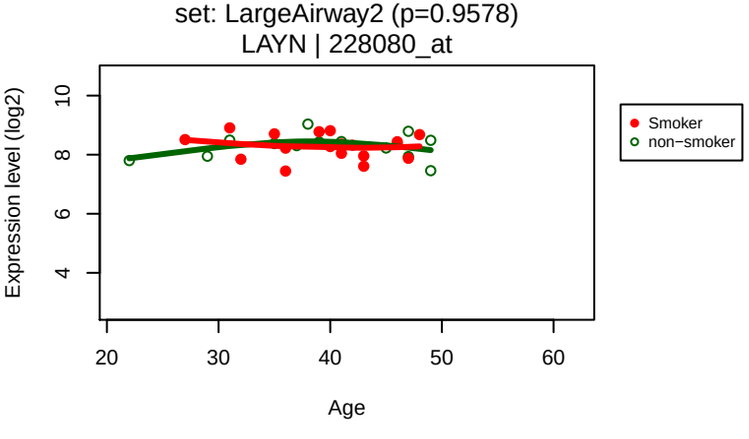

Gene expression ratio (smokers/nonsmokers)
gene: LAYN | 228080_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0605
p=0.0761
p=3e−04
p=0.0381
p=0.9578
p=0.4997
p=0.2933
p=0.819
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

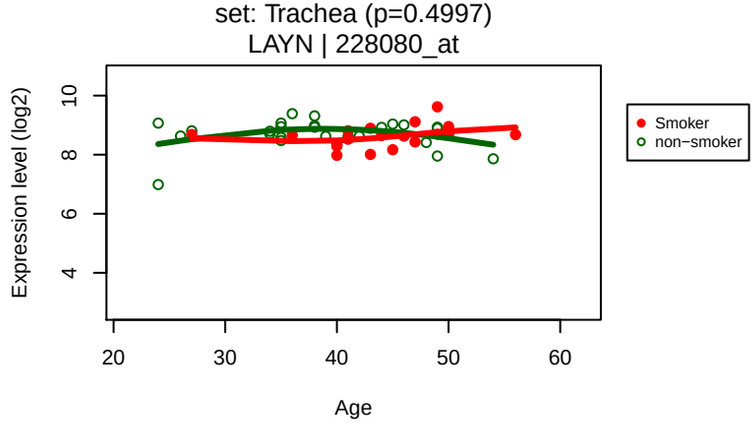

Gene expression ratio in Males
gene: LAYN | 228080_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1337
p=0.9464
p=0.7745
p=0.817
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LAYN | 228080_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3821
p=0.9543
p=0.4982
p=0.5507
SmallAirway2
LargeAirway2
Trachea
Macrophage2

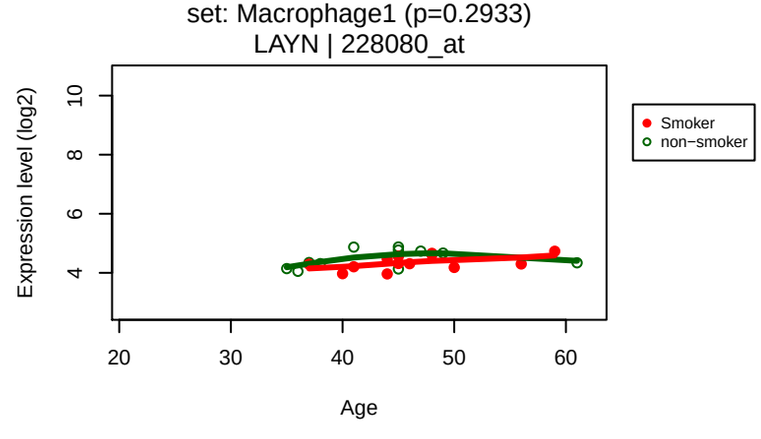

set: SmallAirway1 (p=0.0605)
LAYN | 228080_at
Expression level (log2)
Age
Smoker
non−smoker

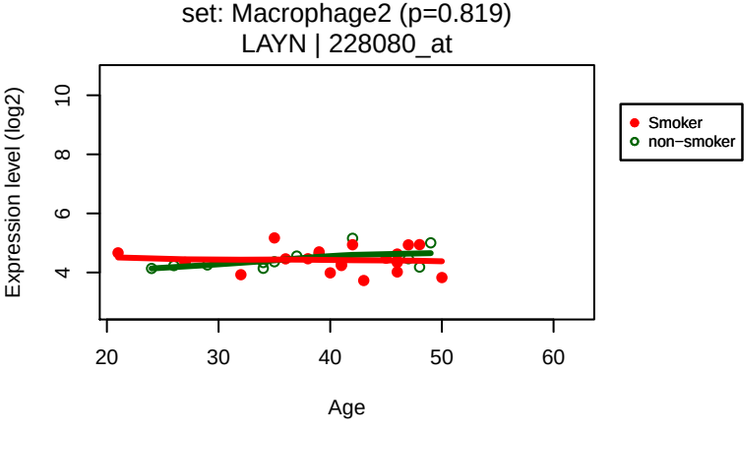

set: SmallAirway2 (p=0.0761)
LAYN | 228080_at
Expression level (log2)
Age
Smoker
non−smoker

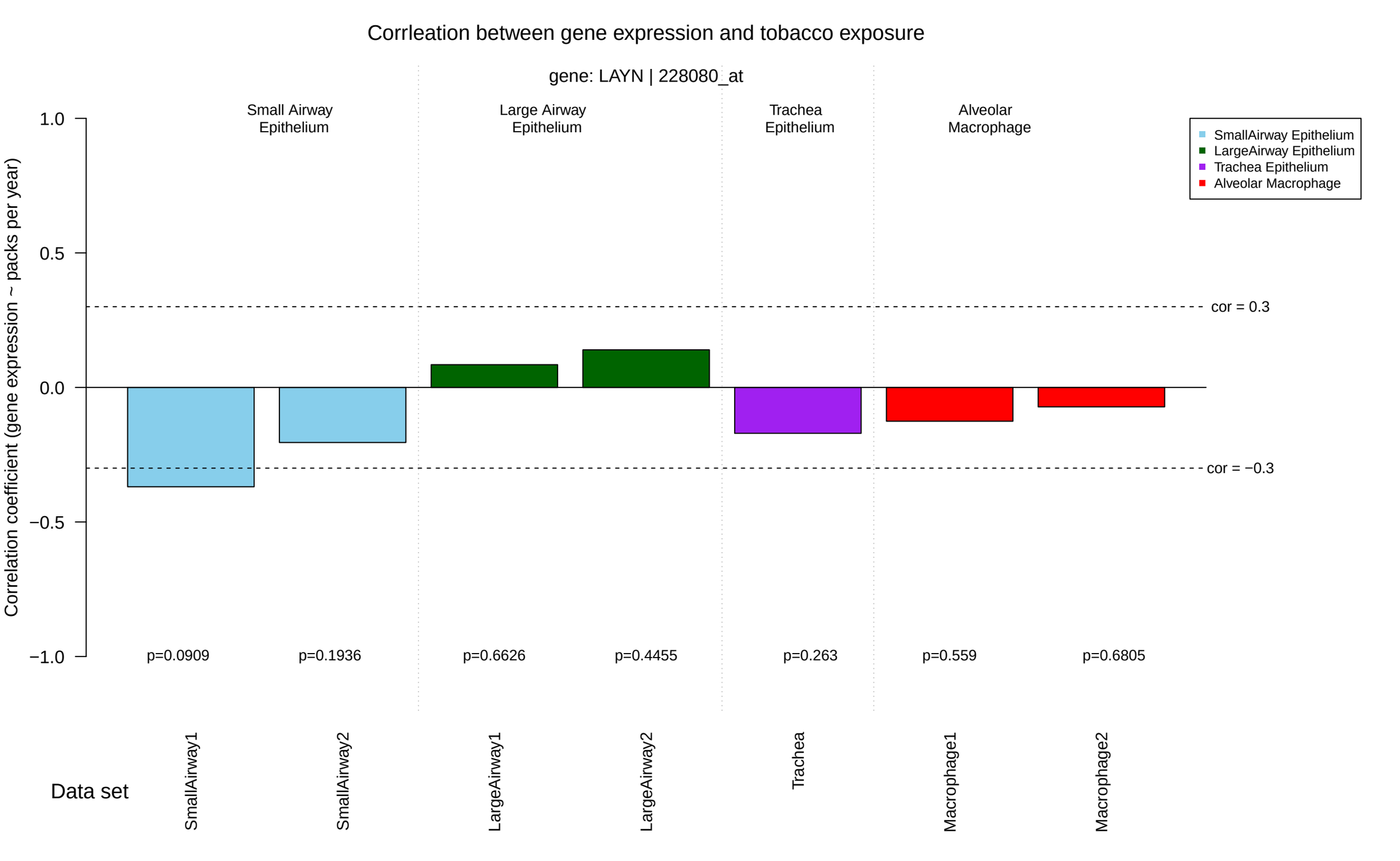

set: LargeAirway1 (p=0.0381)
LAYN | 228080_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9578)
LAYN | 228080_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4997)
LAYN | 228080_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2933)
LAYN | 228080_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.819)
LAYN | 228080_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LAYN | 228080_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0909
p=0.1936
p=0.6626
p=0.4455
p=0.263
p=0.559
p=0.6805
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2